

Cambridge (CIE) IGCSE Additional Maths Paper 2: Calculator (Set A)

Sunday 5 October 2025

Afternoon (Time: 2 hours 0 minutes)

Total marks

/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- Answer all questions.
- You must show all necessary working clearly.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- For π , use either your calculator value or 3.142.

Materials

- You should use a scientific calculator where appropriate.
- [List of formulas](#)

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or visit the mock exams landing page for this course



1 (a) A curve has the equation $y = 5x^2 - 10x + 9$.

Write the equation in the form $y = p(x + q)^2 + r$ where p , q and r are constants to be found.

(2 marks)

(b) Hence find the coordinates of the turning point on the curve.

(1 mark)

(c) Find the range of the function $f(x) = 5x^2 - 10x + 9$ where $x \in \mathbb{R}$.

(1 mark)

(d) Explain why the inverse function $f^{-1}(x)$ does not exist.

(1 mark)

2 Solve the following equation.

$$2|x - 1| \leq |2 - x|$$

(4 marks)

3 Find $\int \frac{1}{(7x + 4)^m} dx$ in the following cases.

(a) $m = 2$

(b) $m = 1$

(4 marks)

- 4 (a) Find the first 3 terms in the expansion of $\left(2 + \frac{k}{2}\right)^8$ in ascending powers of k .

Simplify the coefficient of each term.

(2 marks)

- (b) Hence find the first three terms in ascending powers of p in the expansion of $\left(2 + \frac{p - p^2}{2}\right)^8$.

Simplify the coefficient of each term.

(2 marks)

- (c) Find the term independent of x in the expansion of $\left(ax^2 - \frac{1}{x^2}\right)^6$ where $a \neq 0$.

Give your answer in terms of a .

(3 marks)

- 5 A curve is given by $y = \ln(1 + x^4)$.

Use calculus to find the approximate change in y as x increases from -1 to $-1 + k$ where k is small.

(4 marks)

6 (a) A 4-digit number is created where each digit must be greater than or equal to 3. The same digit must not appear more than once.

(i) Find the number of 4-digit numbers that can be created.

(ii) Find the number of 4-digit numbers that can be created that are less than 9000.

(3 marks)

(b) A shelf holds 5 horror books, 6 romance books and 4 travel books. I choose six books from the shelf to read.

Find the number of selections possible in the following cases:

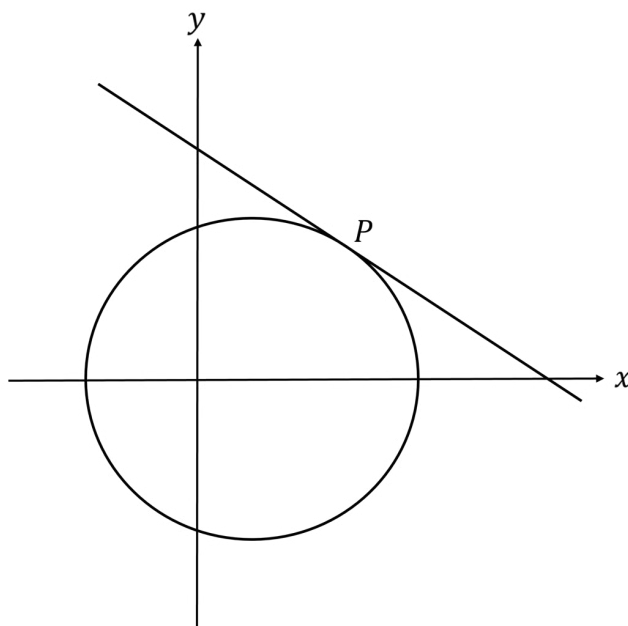
(i) I choose an equal number of each type of book,

(ii) I choose 4 horror books and at least 1 travel book.

(5 marks)

7 (a) The point $P(8, 8)$ lies on the circle $(x - 2)^2 + y^2 = 100$ with centre C .

The circle and the tangent to the circle at P are shown below.



Find the equation of the tangent to the circle at P , giving your answer in the form $y = mx + c$.

(4 marks)

(b) The tangent intersects the y -axis at the point Q .

Find the area of triangle PCQ .

(4 marks)

8 (a) A particle travels along a straight line with a velocity, $v \text{ ms}^{-1}$, given by

$$v = 6\left(t - \frac{1}{2}\right)^2 + \frac{5}{2}$$

for $t \geq 0$ where t is time in seconds. The particle has an initial displacement of -1 metres.

Find the acceleration after 2 seconds.

(4 marks)

(b) Find the displacement after 2 seconds.

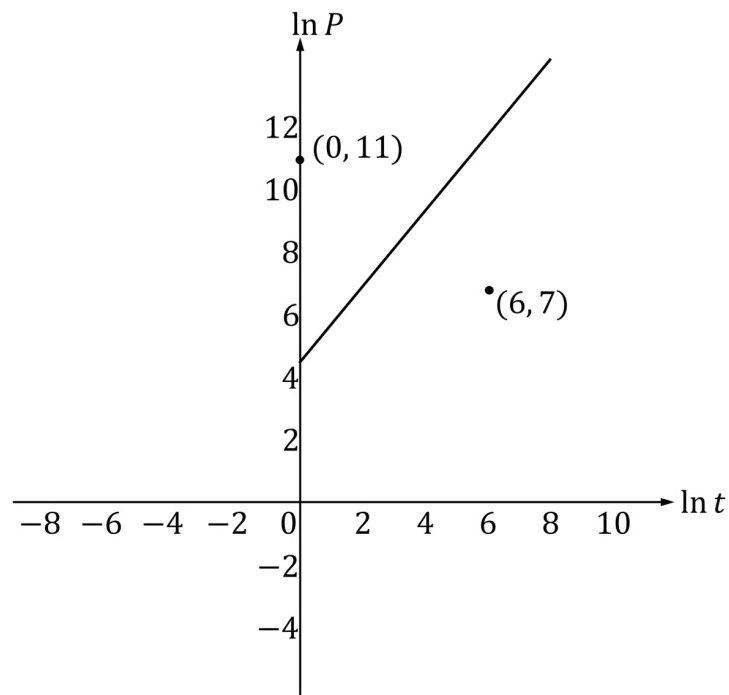
(4 marks)

(c) Explain why a velocity-time graph and a speed-time graph will always be the same for this particle.

(1 mark)

9 A mathematical model is used to predict the population of an island, P , over time, t years.

A plot of $\ln P$ against $\ln t$ gives a straight line that is the **perpendicular bisector** of the points $(0, 11)$ and $(6, 7)$, as shown.

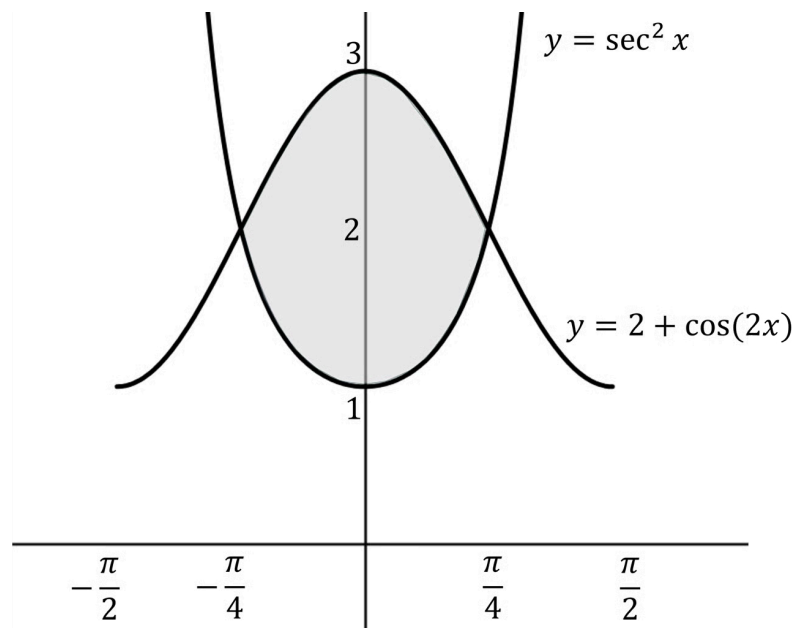


Find and simplify a formula for P in terms of t .

(6 marks)

10 Throughout this question, x is measured in radians.

The curves $y = \sec^2 x$ and $y = 2 + \cos(2x)$ intersect at the points $\left(-\frac{\pi}{4}, 2\right)$ and $\left(\frac{\pi}{4}, 2\right)$, as shown.



Find the exact area of the shaded region enclosed.

(6 marks)

11 (a) In an arithmetic sequence, the 20th term is 10% of the sum of the first 25 terms.

The sum of the 20th term and the 21st term is 73.

Find the 8th term.

(6 marks)

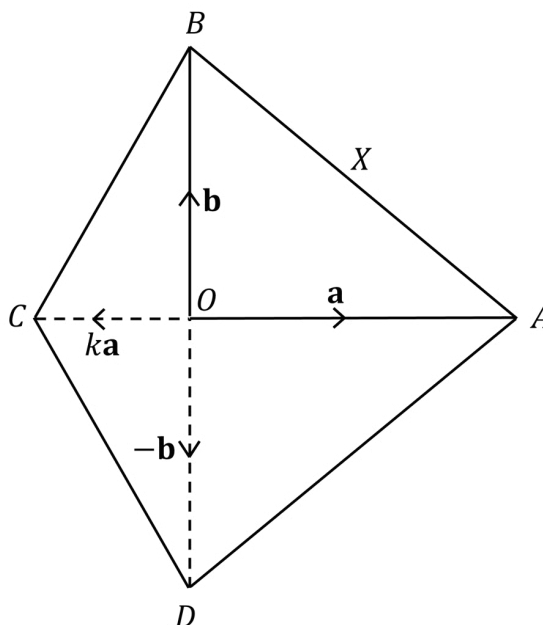
- (b) In a geometric sequence, the sum to infinity is 64 and the sum of the first 7 terms is $\frac{127}{2}$.

Find the ratio of the 4th term to the 7th term, giving your answer in the form $m : 1$.

(6 marks)

- 12 (a)** The kite $ABCD$ has diagonals CA and BD which intersect at O , where $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$, $\vec{OC} = k\mathbf{a}$ and $\vec{OD} = -\mathbf{b}$.

The point X is the midpoint of AB , as shown.



Find

- (i) $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
- (ii) $\vec{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
- (iii) $\vec{CD}$ in terms of k , $\mathbf{a}$ and $\mathbf{b}$

(3 marks)

- (b)** The point Y lies on CD such that $CY:YD = 1:2$.

If $\vec{YX}$ is parallel to $\mathbf{a} + \mathbf{b}$, find k .

(4 marks)